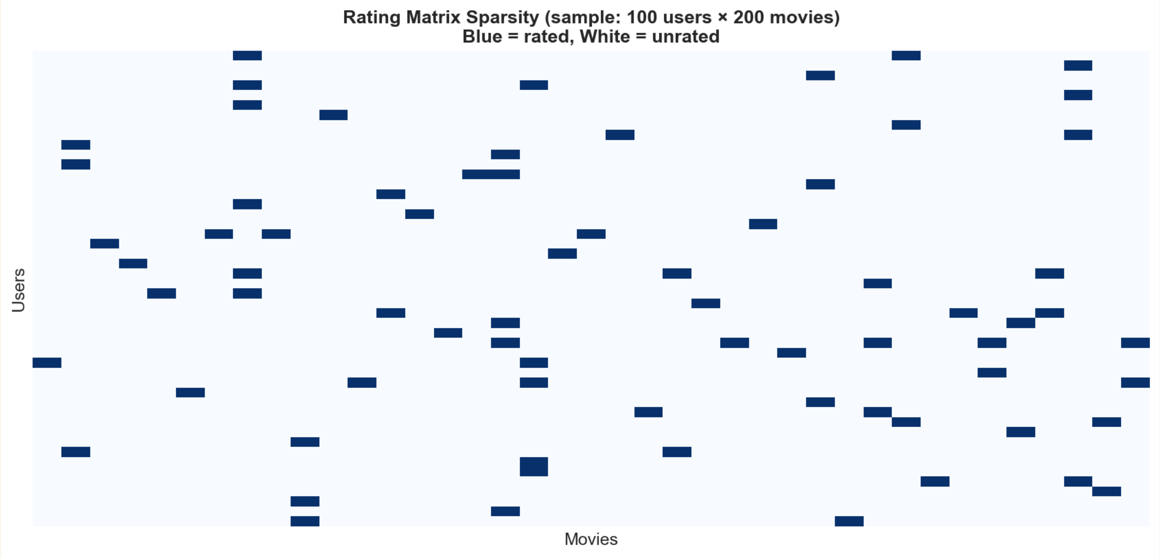
Netflix Recommendation System

Personalized Content Discovery at Scale

Building production-grade collaborative filtering on the **Netflix Prize Dataset** — 100M+ explicit ratings, 480K users, and 17K titles under conditions of extreme sparsity.

01 · PROBLEM DEFINITION

The Challenge & Data Landscape



⚠ **Core Problem — Extreme Sparsity:** Each user rates fewer than 0.01% of the catalogue. Memory-based collaborative filtering (User-CF, Item-CF) fails at this scale due to insufficient overlap between user profiles.

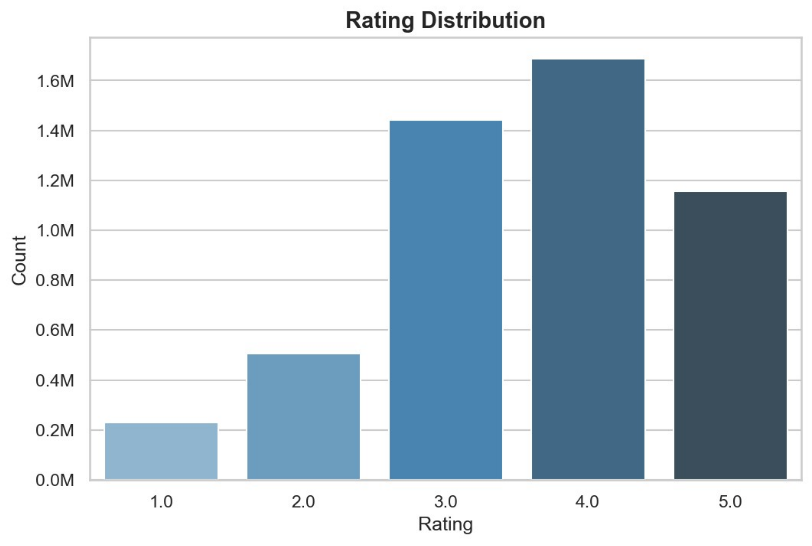
02 · EXPLORATORY ANALYSIS

Exploratory Data Analysis

Key Observations

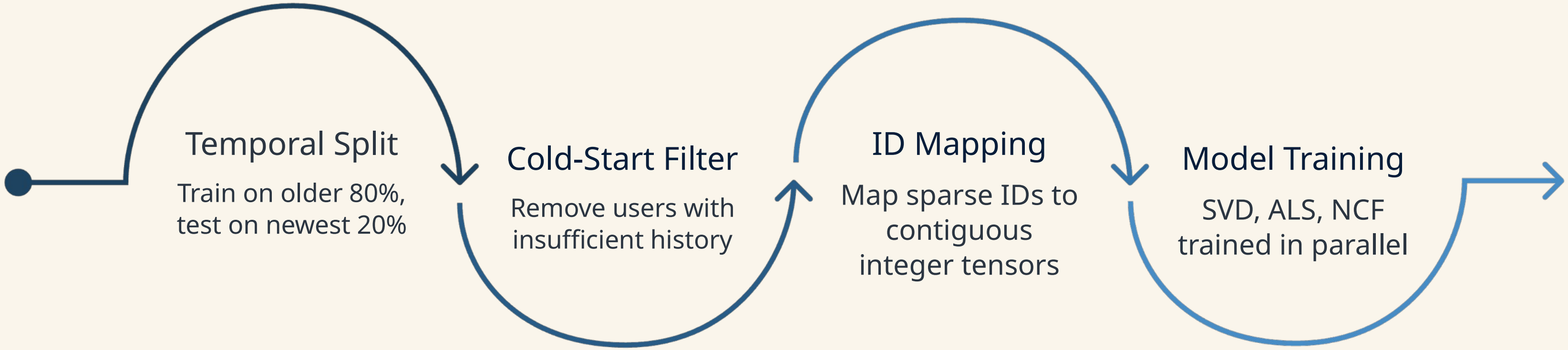
A small fraction of power users generate the majority of ratings, while most users contribute fewer than 20 — a long-tail distribution that biases naive models toward popular titles.

The vast majority of ratings fall between 3 and 5 stars. Models must learn to differentiate "good" from "great," not just "liked" from "disliked."



03 · PIPELINE DESIGN

System Architecture & Pipeline Strategy



Evaluation uses a temporal hold-out — models train on older ratings and are tested on the newest 20%, mirroring real production constraints where only past behaviour is available. Random and leave-one-out splits are also supported. Contiguous integer ID mapping enables efficient sparse-matrix and GPU embedding operations across SVD, ALS, and NCF.

04 · ENGINEERING

Engineering Hurdles & Bug Fixes

Scaling Bottleneck — $O(n^2)$ Similarity

User-CF and Item-CF require ~230 billion pairwise comparisons — computationally infeasible. Pivoted to latent factor models (SVD, ALS) that operate in $O(n \cdot k)$ time.

SVD Type-Cast Bug

SVD scored 0 on ranking metrics due to string-typecasting that forced global average predictions. Root cause: integer ID mapping was silently coerced to strings. Fixed by enforcing explicit integer dtype throughout the pipeline.

ALS RMSE Illusion

Implicit ALS produces unbounded confidence scores, inflating RMSE on the 1–5 scale. Engineered a custom Min-Max scaling layer to normalize outputs to valid star ratings before evaluation.

05 · MODELING

Model Strategy: Three Complementary Approaches

SVD

Singular Value Decomposition

Latent factor model optimized for **RMSE**. Decomposes the user-item matrix into low-rank embeddings capturing taste dimensions.

ALS

Alternating Least Squares

Implicit feedback model optimized for **MAP@10**. Parallelizable and scalable — ideal for Top-K candidate generation.

NCF

Neural Collaborative Filtering

Deep learning architecture replacing the dot product with a multi-layer perceptron to capture **non-linear user-item interactions**.

06 · EVALUATION

Quantitative Evaluation

Dual-Metric Framework

Models were evaluated on two orthogonal objectives:

- **RMSE** — Point-prediction accuracy on the 1–5 star scale
- **MAP@10** — Ranking quality with a ≥ 3.5 relevance threshold

NCF achieved the best RMSE (0.917), with SVD close behind (0.934). ALS dominated every ranking metric — MAP@10, NDCG@10, HitRate@10 and Coverage — by a wide margin, making it the strongest Top-K recommender despite a weaker RMSE.

Metric	SVD	ALS	NCF
RMSE ↓	0.934	1.356	**0.917**
MAP@10 ↑	0.0021	**0.0066**	0.0011
NDCG@10 ↑	0.0036	**0.0115**	0.0031
HitRate@10 ↑	0.0140	**0.0420**	0.0220
Coverage ↑	0.0384	**0.0998**	0.0102

↓ *lower is better* ↑ *higher is better* • *bold = best in row*

Qualitative Insights

The Unobserved Feedback Problem

Several Top-10 samples scored 0% exact-match precision against the 20% temporal test split — yet a qualitative review reveals highly accurate taste mapping. One example recommends Fight Club to a user whose profile centres on acclaimed dramas like The Sopranos and Six Feet Under.

This is a textbook Unobserved Feedback Problem. Under extreme sparsity (>99.99%), the model is penalised for surfacing genuinely relevant, critically acclaimed titles simply because the user never happened to watch them within the test window.

A valid global MAP@10 confirms the model is statistically sound, while these qualitative “failures” prove it generalises to latent taste clusters rather than memorising popularity. In production we would validate such recommendations with online A/B testing — not strict offline exact-match precision.

Final Takeaway

Offline exact-match metrics understate a model that has genuinely learned user taste. For personalised discovery, latent-factor generalisation — confirmed by online A/B testing — matters more than test-set precision.